

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc (BT/MI) Examination May 2018

Course code & Name: MI 751 Microbial Technology

Date: 2/5/2018 Day: Wednesday Time: 1:30 pm To 2:00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions

20

- 1) FDA can direct the industry to take the following step immediately, if a drug fails the QC during inspection
 - (i) seal production facilities
 - (ii) recall the drug
 - (iii) reprocess the drug
 - (iv) all of the above
- 2) The following organism has maximum tendency to move toward a state of lower productivity when used for a fermentation
 - (i) wild type
 - (ii) mutant
 - (iii) natural variant
 - (iv) all of the above
- 3) The following process(es) need(s) to be validated
 - (i) inoculum generation
 - (ii) intermediate fermentation
 - (iii) fermentation
 - (iv) all of the above
- 4) The microbes involved in bioleaching of metal derive their energy from the following
 - (i) glucose
 - (ii) metal oxidation
 - (iii) CO₂
 - (iv) organic acid
- 5) Amylase is used for the hydrolysis of the following
 - (i) nucleic acid
 - (ii) lipid
 - (iii) starch
 - (iv) cellulose
- 6) The following ions is a precursor For the production of vitamin B₁₂
 - (i) cobalt
 - (ii) chromium
 - (iii) calcium
 - (iv) none of the above
- 7) The following proteolytic enzyme is used in detergents as they function at high pH
 - (i) trypsin
 - (ii) chymotrypsin
 - (iii) subtilisins
 - (iv) all of the above

- 8) The following organism is used for the production of Astaxanthin
 - (i) *Aspergillus niger*
 - (ii) *Penicillium oxalicum*
 - (iii) *Candida albicans*
 - (iv) *Xanthophyllomyces dendrorhous*
- 9) Reductive amination of α -ketoacids to produce L-amino acids is commercially viable due to the following
 - (i) recycling of cofactor
 - (ii) high efficiency of enzyme
 - (iii) low affinity for substrate
 - (iv) all of the above
- 10) In the following system of starter culture production, air is sterilized using a combination of heat and cotton wool filtration
 - (i) Jones
 - (ii) tetra pack
 - (iii) Lewis
 - (iv) all of the above
- 11) Food containing *Bifidobacterium longum* is an example of the following
 - (i) probiotic
 - (ii) prebiotic
 - (iii) healthy
 - (iv) synbiotic
- 12) Addition of the following compound for the production of benzyl penicillin, is an example of mutasynthesis
 - (i) benzene
 - (ii) phenyl acetic acid
 - (iii) acetic acid
 - (iv) benzoic acid
- 13) A following is mandatory for organic synthesis using microorganisms or enzymes
 - (i) chiral C
 - (ii) C forming aromatic ring
 - (iii) chiral H
 - (iv) C forming aliphatic chain
- 14) The following protected system of starter culture production, does not require completely enclosed vat for sterilization and cooling of the growth medium
 - (i) microbiologically
 - (ii) mechanically
 - (iii) chemically
 - (iv) physically
- 15) A triggering signal is needed for production of the following
 - (i) citric acid
 - (ii) amino acid
 - (iii) antibiotics
 - (iv) vitamin
- 16) C:N ratio should be between the following for the production of SCP
 - (i) 7: 1 to 10:1
 - (ii) 7: 1 to 15:1
 - (iii) 10:1 to 12:4
 - (iv) 10:1 to 12:4
- 17) The following enzyme introduces one or two oxygen atoms into their substrates and is highly regiospecific
 - (i) Oxygenases
 - (ii) oxidases
 - (iii) peroxidases
 - (iv) dehydrogenase
- 18) The concentration of Mn^{+2} and Mg^{+2} should be kept low in the fermentation medium the production of the following
 - (i) vitaminB₁₂
 - (ii) ethanol
 - (iii) citric acid
 - (iv) lactic acid
- 19) Biocontrol agent protect the plants by _____ the number of pest
 - (i) reducing
 - (ii) increasing
 - (iii) stabilizing
 - (iv) equalizing
- 20) Use of defined medium in a fermentation process increases the cost of the following
 - (i) downstream processing
 - (ii) fermentation process
 - (iii) inoculum generation
 - (iv) all of the above

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc (BT/MI) Examination May 2018

Course code & Name: MI 751 Microbial Technology

Date: 2/5/2018 Day: Wednesday Time: 2:00 pm To 4:30 pm Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I 20

Q – II A Discuss briefly any two of the following: 6

- (i) The criteria to be considered for the validation of fermentation
- (ii) Inoculum generation in a fermentation process
- (iii) Importance of CGMP

Q – II B Attempt any two of the following: 4

- (i) Give an overview of a fermentation process
- (ii) Justify: Though the spores produced by some bacteria are ideally suited to long-term storage, are not ideal for a MCB.
- (iii) Write a short note on tiered seed banking system

Q – III A Discuss briefly any two of the following: 6

- (i) Biomining of gold
- (ii) Conversion of Penicillin G to Ampicilin as an example of semisynthetic antibiotic
- (iii) Differentiate between organic synthesis and biotransformations

Q – III B Attempt any two of the following: 4

- (i) **Discuss briefly:** screening strategies for biocatalyst
- (ii) **Write a short note on;** Biotransformation of steroids
- (iii) **Justify:** The microorganisms involved in biomining are acidophiles and resistant to metal ions

SECTION - II 30

Q – IV A Discuss briefly any two of the following 6

- (i) *In Situ* recovery technology to overcome butanol toxicity during fermentation
- (ii) Selection of C sources for ethanol fermentation
- (iii) Manipulations with Key Genes of the Aspartate Pathway for L-Lysine over-production

Q – IV B	Discuss briefly any one of the following	3
(i)	Characteristics of <i>Bacillus subtilis</i> used by DSM Nutritional Products for the over production of riboflavin	
(ii)	Cyanocobalamine fermentation	
Q – IV C	Justify any one of the following	1
(i)	Production medium for glutamic acid fermentation is supplemented with low concentrations of biotin	
(ii)	Vitamin B is added to the production medium for lactic acid fermentation	
Q – V A	Discuss briefly any two of the following	6
(i)	Fermentative production of penicillin	
(ii)	Food Grade Pigments produced by Yeasts	
(iii)	The ‘signature’ genes/enzymes’ that could be used for the screening of secondary metabolite producing micro-organism	
Q – V B	Discuss briefly any two of the following	4
(i)	Carbon catabolite regulation for the regulation of secondary metabolite production with example	
(ii)	Approaches to isolate microbes that produce new bioactive small molecules	
(iii)	Production of fungal lovastatin	
Q – VI A	Discuss briefly any two of the following	6
(i)	Mechanically protected systems for the production of starter cultures	
(ii)	The characteristics of a probiotic strain	
(iii)	The characteristics of microorganisms for SCP and its cultivation	
Q – VI B	Attempt any two of the following	4
(i)	List the types of biofertilizers and explain the importance of each in agriculture	
(ii)	Discuss briefly : The government regulations for used of biocontrol agents	
(iii)	Write a short note on SCO	